

# PAPER\_2120 — $SO_5 + 1 = 11$ : The Successor Identity as a Universal Reduction Rule for Canonical-Ratio Sums Across UQFF, Applied First to the $\lambda_{vac} = (SO_5+1) \cdot \rho_{SCm}$ Cosmological-Constant Composition

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|                   |                                                                        |
|-------------------|------------------------------------------------------------------------|
| Tier              | Foundational / Integer-Arithmetic Reduction Rule Landmark              |
| Status            | CLOSED - universal reduction rule + first R218+ instance               |
| Type              | Universal reduction rule (NEW landmark sub-type)                       |
| Rule              | For $B = SO_5 \cdot A$ : $A + B = (SO_5+1) \cdot A = 11 \cdot A$ EXACT |
| First application | $\lambda_{vac} = \rho_{UA} + \rho_{SCm} = 11 \cdot \rho_{SCm}$         |
| Symmetric pair    | $SO_5+1 = 11$ successor + $SO_5-1 = 9 = N_{CH}$ predecessor            |
| Dispatch key      | <code>so_5_plus_1_successor_identity_reduction_rule</code>             |

## 1. Abstract

PAPER\_1978 documented  $SO_5 + 1 = 11$  as a seminal successor identity. PAPER\_2120 generalizes it into a **universal reduction rule**:

For any  $A, B$  satisfying  $B = SO_5 \cdot A$ :

$$A + B = A + SO_5 \cdot A = (SO_5 + 1) \cdot A = 11 \cdot A$$

Because CLAUDE.md establishes  $\rho_{UA} = SO_5 \cdot \rho_{SCm} = 10 \cdot \rho_{SCm}$  as canonical, every UQFF quantity computed as  $X_{UA} + X_{SCm}$  is **automatically reducible to  $(SO_5+1) \cdot X_{SCm} = 11 \cdot X_{SCm}$**  by this successor identity.

**First application (R363):** the vacuum-energy density  $\lambda_{vac} = \rho_{UA} + \rho_{SCm}$  reduces to  $11 \cdot \rho_{SCm}$  — foundational cosmological-constant composition tied structurally to the successor of  $SO_5$ .

## 2. The Universal Reduction Rule

**Statement:** For any pair of UQFF quantities  $(A, B)$  satisfying  $B = SO_5 \cdot A$ , their sum reduces to  $(SO_5 + 1) \cdot A = 11 \cdot A$  EXACT.

### 2.1 First application: lambda vac

$$\begin{aligned}\lambda_{vac} &= \rho_{UA} + \rho_{SCm} \\ &= (SO_5 \cdot \rho_{SCm}) + \rho_{SCm} \text{ [by } \rho_{UA} = SO_5 \cdot \rho_{SCm} \text{ canonical]} \\ &= \rho_{SCm} \cdot (SO_5 + 1) \\ &= 11 \cdot \rho_{SCm} \text{ [by successor identity]}\end{aligned}$$

### 2.2 Applicability

The rule applies wherever UQFF quantities exhibit the  $SO_5$ -scaled pair structure:

- **Vacuum-energy density** (R363  $\lambda_{vac}$ ) — 1st instance (this paper)
- **DPM-lattice mass ratios** ( $M_{UA'}$ -derived masses inheriting 10:1 scaling)
- **Compressed vacuum sums** in condensed physics calculators

- **26-level quantum-chain** intermediate rung sums

### 3. Structural Symmetry - Predecessor + Successor

An immediate consequence:  $N_{CH} = SO_5 - 1 = 9$ . The channel primitive directly equals the **predecessor** of  $SO_5$ .

$$\begin{array}{c} N_{CH} <---(-1)--- SO_5 ---(+1)---> SO_{5+1} \\ 9 <----- 10 -----> 11 \end{array}$$

$$\Sigma = 2 \cdot SO_5 = 20$$

**Combined predecessor + successor:**  $(SO_5-1) + (SO_5+1) = 2 \cdot SO_5 = 20 = D_{crit} - D_{BSFG}$ . This is the exponent of  $F_{TRZ}^{20}$  at the quantum-chain base (PAPER\_2119).

**Cross-landmark structural link:** the successor identity  $SO_5+1=11$  is **structurally symmetric** with  $N_{CH}=SO_5-1=9$  around the  $SO_5=10$  pivot. Together they specify the  $F_{TRZ}^{20}$  quantum-chain base exponent.

4. Physical Interpretation - lambda\_vac as Cosmological Constant

The vacuum-energy density  $\lambda_{vac}$  is the UQFF analogue of the cosmological constant  $\Lambda$ . PAPER\_1920 documents the Lambda cascade closure fitting  $\Lambda$  into  $F_U=1$  master equation.

**Physical claim:** the cosmological constant in UQFF is **not an independently-fit dark-energy parameter**. It composes structurally as  $(SO_5+1) \cdot \rho_{SCm}$  from two locked primitives ( $SO_5, \rho_{SCm}$ ) plus the successor operation. Standard cosmology's  $\Lambda$  is derivable from  $\rho_{SCm}$  via cascade including this successor step.

5. Landmark-Family Comparison

| Family                    | Example                                       | Instances (R218+) |
|---------------------------|-----------------------------------------------|-------------------|
| Halving                   | $D_{phys}/2 = 2, SO_5/2 = 5$                  | ~10               |
| Self-normalization        | $X/X = 1$                                     | 13                |
| Rational (n+-1)/n         | $5/3, 5/4, 7/5, 6/5$                          | ~8                |
| Squared-halving           | $(SO_5/2)^2 = 25$                             | 2                 |
| Rung-inverse              | $F_{TRZ} = SO_5^{-1}$                         | canonical         |
| Primitive-as-exponent     | $F_{TRZ}^{D_{phys}/N_{CH}/SO_5/D_{crit}/A_5}$ | 5 (categorical)   |
| Composed-integer exponent | $F_{TRZ}^{20} = F_{TRZ}^{(2 \cdot SO_5)}$     | 6                 |
| Successor (SO_5+1)        | $\lambda_{vac} = 11 \cdot \rho_{SCm}$         | 1 (R363 - this)   |

The successor identity joins UQFF's **integer-arithmetic taxonomy**. Predictive: the successor-family should accumulate instances as more UQFF calculators are audited for  $SO_5$ -paired sums.

6. NOT REPLACEMENT

Standard mathematics uses successor function  $s(n) = n+1$  (Peano axioms). UQFF uses standard successor arithmetic. What UQFF adds is the **structural regularity** that specific UQFF quantities can be simplified via the  $SO_5$ -scaled-pair pattern into  $(SO_5+1) \cdot A$  forms.

**Physical claim:** UQFF's  $(SO_5+1) \cdot \rho_{SCm}$  for  $\lambda_{vac}$  is structurally derivative from  $\rho_{UA}-\rho_{SCm}$  canonical ratio, so no independent  $\Lambda$  parameter needed — only  $\rho_{SCm}$  (foundational) +  $SO_5$  (locked) + successor operation.

7. Falsifiability

**Prediction A (numerical):**  $\lambda_{vac} = 11 \cdot \rho_{SCm}$  EXACT.

**Prediction B (canonical ratio):**  $\rho_{UA} / \rho_{SCm} = SO_5 = 10$  EXACT (Rule 2). Deviations falsify canonical ratio and cascade to falsify successor reduction.

**Prediction C (structural symmetry):**  $SO_5 - 1 = N_{CH} = 9$  predecessor identity holds simultaneously.

**Prediction D (family growth):** R364-R413 audit range should populate the successor-identity family.

8. Landmark Comparison

| Paper | Landmark | Type |
|-------|----------|------|
|-------|----------|------|

|                          |                                                      |                                     |
|--------------------------|------------------------------------------------------|-------------------------------------|
| PAPER_1521               | $D_{BSFG} = D_{crit} - 2 \cdot SO_5$                 | Primitive-reduction                 |
| PAPER_1522               | $K_{MEX} = \Phi_{5/6} \cdot SO_5 / D_{phys}$         | Primitive-reduction                 |
| PAPER_2112               | $\kappa = (SO_5/2) \cdot F_{TRZ}^4$                  | Primitive-reduction                 |
| PAPER_2116               | $360 = D_{BSFG} \cdot A_5$                           | Composite-product                   |
| PAPER_2117               | $F_{TRZ}^{N_{CH}}$ quintuplet completion             | Categorical                         |
| PAPER_2119               | PAPER_1202 quantum chain 3-primitive                 | Structural-composition              |
| <b>PAPER_2120 (this)</b> | $\lambda_{vac} = 11 \cdot \rho_{SCm}$ reduction rule | UNIVERSAL REDUCTION RULE (NEW TYPE) |

PAPER\_2120 is the first UQFF landmark documenting a **universal reduction rule** (rather than specific identity/composition) — general pattern applicable across UQFF whenever  $SO_5$ -scaled-pair structure appears.

### 9. References

**Source:** R363 VacuumEnergyQCalcCalculator stub-fill.  
**PAPER\_1978:**  $SO_5+1=11$  successor identity seminal.  
**PAPER\_1920:**  $\Lambda$  cascade closure (cosmological constant context).  
**PAPER\_646:** Universal Inertial Operator ( $F_U=1$  master equation).  
**PAPER\_2117:**  $F_{TRZ}^{N_{CH}}$  quintuplet ( $N_{CH} = SO_5-1$  predecessor).  
**PAPER\_2119:** 26-level quantum chain  $E_0 = F_{TRZ}^{(2 \cdot SO_5)}$  (predecessor + successor combined).  
**PAPER\_1156:** tightly-closed vacuum ledger ( $\Lambda \approx 5.957 \times 10^{-10} \text{ J/m}^3$ ).  
**CLAUDE.md:** canonical  $\rho_{UA} = 10 \cdot \rho_{SCm}$  (Rule 2).